HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Assisted in the root cause analysis of **process deviations** by analyzing **scientific data** and providing data-driven rationale.
- Participated in **cross-functional teams** to investigate technical and laboratory issues impacting process performance.
- Contributed to the creation of knowledge-based reports summarizing study conclusions, investigations, and identified **process improvements**.
- Produced and maintained accurate laboratory records in accordance with site procedures and industry guidelines.
- Developed **Python scripts** to automate data extraction from various sources, enabling faster report generation for analytical teams.
- Designed and implemented **Java Spring Boot** REST APIs to support data exchange between enterprise systems, ensuring seamless integration.
- Containerized applications using **Docker** and managed deployments on **AWS EC2** instances, improving application portability and scalability.

SKILLS

Laboratory & Process Sciences – Process Validation, Root Cause Analysis, Scientific Data Analysis, Process Improvement, Knowledge-Based Reporting, Laboratory Record Keeping

Programming Languages – Python, Java, SQL, R

Frameworks & Libraries – Spring Boot, pandas, NumPy, scikit-learn

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Databases – PostgreSQL, MySQL, Oracle

BI & Analytics Tools – Power BI, Tableau, DAX, Power Query

PROJECTS

Automated Data Pipeline for Process Analysis

- Developed a **Python ETL pipeline** to ingest data from laboratory instruments and operational databases, consolidating it for analysis.
- Implemented data validation checks within the pipeline to ensure data integrity, reducing errors in downstream reporting by approximately **15%**.
- Built automated reports using **pandas** and scheduled their execution via cron jobs on a **Linux** server, providing timely insights.

JMS Message Processing and Monitoring System

- Engineered a **Java Spring Boot** application to consume and process messages from a **JMS** queue, facilitating transaction monitoring.
- Integrated **JPA** with a **PostgreSQL** database to persist transaction details and operational logs, supporting reconciliation workflows.
- Configured **Splunk** alerts for critical transaction failures, reducing the mean time to detect issues by **20%**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)